

L1.6 Trig Derivatives & the Chain Rule

Work in your group. Check your answers against the Quarto tonight.

The six trig derivatives

1. For each, say whether the problem **forces radians**, **forces degrees**, or leaves you **free to choose** — and give the reason.

(a) $\frac{d}{dx}(\sin x)$

(b) reading $\sin 30^\circ$ off the unit circle

(c) $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}$

(d) finding an angle in a right triangle with legs 3 and 4

2. Fill in the left column from memory. Then **generate** the right column — do not recall it separately.

function	derivative	co-function	derivative
$\sin x$		$\cos x$	
$\tan x$		$\cot x$	
$\sec x$		$\csc x$	

If you know the derivative of a trig function, how do you find the derivative of its co-function?

3. Derive $\frac{d}{dx}(\tan x)$ **from the quotient rule**, starting from $\tan x = \frac{\sin x}{\cos x}$.

4. Differentiate.

(a) $y = x^2 \sin x$

(b) $y = \frac{\tan x}{x}$

5. Evaluate.

(a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{3x}$

(b) $\lim_{x \rightarrow 0} \frac{\sin x}{\tan x}$

(c) $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{5\theta}$

Differentials

6. Find dy in each case, at $x = 1$ with $dx = 0.1$. Then say what y changes to.

(a) $y = x + 1$

(b) $y = x^2$

(c) $y = \sin x$ (*this one at $x = 0$*)

The input nudge is $dx = 0.1$ every time. Why doesn't dy always come out the same?

The chain rule

7. $y = \sin(x^2)$, so the inner function is $h = x^2$ and the outer is $f = \sin h$. Start at $x = 1.5$ and nudge by $dx = 0.1$. Fill in the table. Use radians.

	line	at	nudge	after
①	x	1.5	$dx = 0.1$	
②	$h = x^2$		$dh = 2x dx =$	
③	$f = \sin h$		$df = \cos(h) dh =$	

Now divide the last line by dx . What have you written?

8. **Decompose only. Differentiate nothing.** Name the outer function and the inner function.

$\sin(x^3)$	$\sin^3 x$	$\sqrt{4x+1}$	$\tan(2x)$	$(x^2+x)^{10}$
outer _____	outer _____	outer _____	outer _____	outer _____
inner _____	inner _____	inner _____	inner _____	inner _____

9. Differentiate.

(a) $y = (4x - 3)^6$

(b) $y = \sin(5x)$

(c) $y = \cos^4 x$

(d) $y = \sec(x^2 + 1)$

10. $f(x) = \sqrt{x^2 + 1}$. Find $f'(x)$, then find $f'(2)$.

The chain rule with the product and quotient rules

Stop. What is the outermost operation — a product, a quotient, a power, or a trig function?

11. Name the outermost operation, then the rule for each piece.

function	outermost	rule for each piece
$(2x + 1)^2 x^3$		
$\frac{\sin x}{x^2 + 1}$		
$\sin(x^2 + 1)$		
$(x^2 \sin x)^4$		

12. $y = (2x + 1)^2 x^3$. Differentiate it, then expand and simplify.

13. **BOARDS** Two problems at the boards.

- (a) For $f(x) = x^3$, find the differential $df = d(x^3)$. Then use it to say how much the output changes as x goes from 1 to 1.2.
- (b) Differentiate $g(x) = \frac{\sec(\tan x)}{\tan(\sec x)}$.